

AH-LoRA: Heterogeneous Low-Rank Adapters for Decentralized Multi-Domain Federated Learning

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Abstract

Federated fine-tuning with Low-Rank Adaptation (LoRA) must handle statistical heterogeneity, unequal device budgets, and unreliable central coordination at the same time. We present AH-LoRA, a framework whose implemented core aggregates LoRA adapters in effective-update space rather than factor space. A client sends $\Delta W_i = (\alpha/r_i)B_iA_i$ with a shared scaling constant α ; neighbors mix these matrices and each receiver refactorizes the result to its own rank by truncated SVD. This operation is invariant to the LoRA factor gauge and supports heterogeneous ranks in a serverless graph. We implement Metropolis–Hastings gossip, an oracle two-tier hierarchy, heterogeneous-rank and error-feedback protocols, direction-aware signatures, online domain discovery, and an adaptive-rank controller. On a real CIFAR-100 benchmark with 15 clients, five known domains, frozen ImageNet ResNet-18 features, 50 rounds, and three seeds, the oracle hierarchy reaches $68.98 \pm 0.95\%$ personalized accuracy at uniform rank 16, compared with $42.30 \pm 1.39\%$ for flat gossip and $23.20 \pm 0.29\%$ for centralized FedAvg under the same personalized evaluation. The online discovery mixer infers five groups with stage-10 scoring ARI 1.00 for all three seeds (stage-20 ARI 1.00/0.918/1.00 under coordinator-visible full-state observation). The single-seed Project 1 rank controller saves 93.75% of measured FLOPs but selects rank 2 throughout and loses accuracy on three of five tasks, so it is an implementation baseline rather than a calibrated allocator. We also regenerate the Project 2 aggregation-weight calibration on five real datasets; its predictive fit is weak and should be treated as exploratory. The resulting artifact is a reproducible benchmark and implementation record with explicit limits on generalization.

1 Introduction

Large pretrained models are commonly specialized with LoRA [5], which trains small factors $B \in \mathbb{R}^{d_o \times r}$ and $A \in \mathbb{R}^{r \times d_i}$ while freezing the backbone. Existing federated LoRA methods address parameter efficiency and, in some cases, rank heterogeneity [1, 2, 4, 7]; decentralized optimization provides serverless gossip protocols [3, 6]. These lines of work leave a practical intersection underspecified: clients may have different ranks, different domains, and no trusted aggregation server.

LoRA factors are not identifiable: $(BQ, Q^{-1}A)$ represents the same update as (B, A) for every invertible Q . Factor averaging therefore depends on arbitrary coordinates and is undefined for unequal ranks. AH-LoRA defines the communicated object in the common matrix space $\Delta W_i = (\alpha/r_i)B_iA_i$ with shared α and refactorizes every mixed update with an SVD. This primitive is local, rank-agnostic, and compatible with symmetric doubly-stochastic mixing.

Our contribution is deliberately scoped: a tested effective-update merge and Metropolis–Hastings gossip implementation; a reproducible CIFAR-100 benchmark; heterogeneous-rank, factor zero-padding, and error-feedback ablations; offline and online direction-aware discovery; an auditable adaptive-rank controller; and a real-data regeneration of the Project 2 calibration pipeline.

Neighborhood-local discovery, calibrated rank adaptation, privacy guarantees, and broad multimodal generalization remain open.

2 Problem Formulation

Let clients $i \in \{1, \dots, N\}$ communicate over an undirected graph G . Client i owns private data from domain $d(i)$ and maintains a LoRA update ΔW_i . Ranks r_i may differ but must satisfy local resource ceilings. A symmetric doubly-stochastic mixing matrix W preserves the network mean. We report personalized accuracy (each client on its held-out shard) and consensus accuracy (a reference-rank update on the union of test shards), plus worst-domain accuracy, the best-minus-worst domain gap, consensus distance, residual energy, and effective versus operational communication counts. The idealized objective is

$$\min_{\{\Delta W_i\}} F = \frac{1}{N} \sum_{i=1}^N f_i(W_0 + \Delta W_i) \quad \text{s.t.} \quad \text{rank}(\Delta W_i) \leq r_i, \quad (i, j) \in E \text{ for every transmitted pair.} \quad (1)$$

The reported personalized score is $A_{\text{pers}} = N^{-1} \sum_i \text{Acc}_i(\mathcal{T}_i)$; the consensus adapter is $\Delta W_{\text{con}} = \mathcal{C}_R(N^{-1} \sum_i \Delta W_i)$ and the fairness gap is $\max_d A_d - \min_d A_d$.

3 Method

AH-LoRA combines three components that can be evaluated independently and then composed: domain-aware weighting (Project 2), gauge-invariant decentralized aggregation (Project 3), and stateful rank selection (Project 1). The following subsections define the operators used in the experiments and make explicit which quantities are observed, communicated, and optimized.

3.1 Domain-aware aggregation weights

For client i , let n_i denote the local sample count, ℓ_i its measured local loss, $q_i = (1 + \ell_i)^{-1}$ its quality score, and λ_i its domain-aware multiplier. Project 2 normalizes the contribution used by the weighted aggregator as

$$a_i = \frac{n_i q_i \lambda_i}{\sum_j n_j q_j \lambda_j}, \quad \sum_i a_i = 1. \quad (2)$$

The fitted quality score is linear in standardized measurements, $s_i = \theta_0 + \theta^\top x_i$. Within each task/round group g , the multiplier is centered before exponentiation and then clipped and renormalized:

$$\lambda_i = \text{renorm}_{[0.5, 1.5], \text{mean}=1}(\exp[\gamma(s_i - \bar{s}_g)]), \quad \bar{s}_g = |g|^{-1} \sum_{j \in g} s_j. \quad (3)$$

The standardized calibration estimates its coefficients by ridge regression,

$$\hat{\theta}_\alpha = \arg \min_{\theta} \{\|X\theta - y\|_2^2 + \alpha \|\theta\|_2^2\}, \quad \hat{\theta}_\alpha = (X^\top X + \alpha I)^{-1} X^\top y, \quad (4)$$

with α selected by leave-one-task-out RMSE. The two calibration forms differ only in the components of x_i ; both are treated as exploratory when predictive fit is weak.

The measured domain signals begin with class proportions $p_{ic} = n_{ic} / \sum_c n_{ic}$. We record entropy, normalized entropy, and an imbalance statistic,

$$H_i = - \sum_{c: p_{ic} > 0} p_{ic} \log p_{ic}, \quad H_i^{\text{norm}} = H_i / \log C, \quad B_i = \frac{\max_c n_{ic}}{\min_{c: n_{ic} > 0} n_{ic}} \left(1 + \frac{z_i}{C}\right), \quad (5)$$

where z_i counts absent classes. With global proportions q_c and $m_c = (p_{ic} + q_c)/2$, the recorded divergences are

$$\text{KL}(p_i \| q) = \sum_c p_{ic} \log \frac{p_{ic} + \varepsilon}{q_c + \varepsilon}, \quad \text{JS}(p_i, q) = \frac{1}{2} \text{KL}(p_i \| m) + \frac{1}{2} \text{KL}(q \| m), \quad (6)$$

and update-distance features are $d_{\cos}(u, \bar{u}) = 1 - u^\top \bar{u} / (\|u\|_2 \|\bar{u}\|_2)$ and $d_2(u, \bar{u}) = \|u - \bar{u}\|_2$. The weighted rank-assignment variant combines normalized signals as

$$u_i = \lambda_c C_i + \lambda_e E_i + \lambda_b B_i, \quad \tilde{r}_i = r_{\min} + \frac{u_i}{\sum_j u_j} (B_{\text{tot}} - N r_{\min}), \quad (7)$$

then snaps to the allowed menu and greedily corrects to the total rank budget; if $\sum_j u_j = 0$, uniform weights are used.

3.2 Gauge-invariant effective-update aggregation

A LoRA adapter with factors $B_i \in \mathbb{R}^{d_o \times r_i}$ and $A_i \in \mathbb{R}^{r_i \times d_i}$ communicates the scaled effective update using a shared scaling constant α

$$\Delta W_i = \frac{\alpha}{r_i} B_i A_i. \quad (8)$$

The P3 runner and all heterogeneous-rank comparisons use this scaled update; the legacy Project 1 FedAvg path retains its historical unscaled $B_i A_i$ aggregation and is reported separately. For neighbors $\mathcal{N}(i)$, client i forms a convex update-space merge,

$$\overline{\Delta W}_i = \sum_{j \in \mathcal{N}(i) \cup \{i\}} w_{ij} \Delta W_j, \quad \sum_j w_{ij} = 1, \quad w_{ij} \geq 0. \quad (9)$$

If $\overline{\Delta W}_i = U \Sigma V^\top$, its rank- r_i representation is obtained from the truncated SVD:

$$\begin{aligned} \mathcal{C}_{r_i}(\overline{\Delta W}_i) &= U_{(:,1:r_i)} \Sigma_{1:r_i} V_{(:,1:r_i)}^\top, \\ B_i &= \frac{U_{(:,1:r_i)} \Sigma_{1:r_i}^{1/2}}{\sqrt{\alpha/r_i}}, \quad A_i = \frac{\Sigma_{1:r_i}^{1/2} V_{(:,1:r_i)}^\top}{\sqrt{\alpha/r_i}}. \end{aligned} \quad (10)$$

The projection is the Eckart–Young minimum-Frobenius-error rank- r_i approximation. It is invariant to the LoRA factor gauge: replacing (B, A) by $(BQ, Q^{-1}A)$ leaves BA unchanged, while entry-wise factor averaging changes in general and introduces cross terms.

Proposition 1 (Factor averaging is gauge dependent). *Replacing one factor pair by $(BQ, Q^{-1}A)$ leaves BA unchanged but changes entry-wise factor averages in general; expansion also introduces cross terms. Thus factor averaging is not a function of represented updates, whereas the ΔW average is.*

3.3 Decentralized mixing, hierarchy, and feedback

With a symmetric doubly-stochastic mixing matrix W^t , client i receives

$$Y_i^t = \sum_{j=1}^N W_{ij}^t \Delta W_j^t + m_i^t, \quad \overline{\Delta W}_i^t = \mathcal{C}_{r_i}(Y_i^t), \quad (11)$$

where $m_i^t = 0$ without error feedback. The discarded residual is retained as

$$m_i^{t+1} = Y_i^t - \mathcal{C}_{r_i}(Y_i^t). \quad (12)$$

Flat gossip uses ring Metropolis–Hastings weights, $W_{ij} = 1/(1 + \max(d_i, d_j))$ on edges and $W_{ii} = 1 - \sum_{j \neq i} W_{ij}$. The oracle two-tier schedule is $W^t = W_{\text{intra}}^t W_{\text{bridge}}^t W_{\text{intra}}^t$ on bridge rounds and $W^t = W_{\text{intra}}^t$ otherwise. It uses known domains only as an oracle diagnostic.

Communication is reported in factor floats rather than matrix entries. If source j has rank r_j and layer dimensions $(d_{o,\ell}, d_{i,\ell})$, its payload is $S_j = \sum_{\ell} r_j(d_{o,\ell} + d_{i,\ell})$ floats. For mixer support $E_t = \{(i, j) : i \neq j, W_{ij}^t > 0\}$, the effective count is

$$F_t = \sum_{(i,j) \in E_t} S_j. \quad (13)$$

3.4 Automatic discovery and adaptive mixing

Online discovery compresses the flattened effective updates using a fixed seeded projection $P \in \mathbb{R}^{64 \times D}$. The normalized signature and its exponential moving average are

$$z_i^t = \frac{P \text{vec}(\Delta W_i^t)}{\|P \text{vec}(\Delta W_i^t)\|_2}, \quad h_i^t = \frac{\beta h_i^{t-1} + (1 - \beta) z_i^t}{\|\beta h_i^{t-1} + (1 - \beta) z_i^t\|_2}. \quad (14)$$

The affinity is $A_{ij} = h_i^\top h_j$. On the selected topology, we construct a nonnegative kernel and project it to a symmetric mixer,

$$K_{ij} = \mathbf{1}\{j \in \mathcal{N}(i) \cup \{i\}\} \exp\left(\frac{A_{ij} - A_{\max}}{\tau}\right), \quad W_0 = \text{Sinkhorn}(K), \quad W = w_{\min} I + (1 - w_{\min}) W_0. \quad (15)$$

The unknown cluster count is selected without labels by $\hat{K} = \arg \max_{k \in \{2, \dots, \min(8, N-1)\}} \text{Silhouette}(1 - A, k)$. If all scores are non-positive, $\hat{K} = 1$ and the soft mixer is retained; labels and ARI/NMI are used only for evaluation.

3.5 Adaptive rank selection

Project 1 estimates the stable rank of a probe gradient G_i and applies a capability floor:

$$s_i = \frac{\|G_i\|_F^2}{\|G_i\|_2^2}, \quad c_i = \frac{\text{index}(b_i)}{M - 1}, \quad r_i^* = q_{\mathcal{R}_i}(\max\{r_{\min}, \gamma c_i R_i^{\max}, \min(s_i, R_i^{\max})\}), \quad (16)$$

where $q_{\mathcal{R}_i}$ snaps to the allowed candidate menu. At round t , optional residual demand ρ_i^t modifies the signal as

$$d_i^t = \text{clip}(s_i^t [1 + \omega \text{clip}(\rho_i^t, 0, 1)], 0, R_i^{\max}), \quad \tilde{d}_i^t = \beta_r \tilde{d}_i^{t-1} + (1 - \beta_r) d_i^t. \quad (17)$$

The stateful controller changes rank only after the configured hysteresis margin and patience conditions are satisfied. This makes rank decisions auditable while allowing capability ceilings to be enforced at every client.

The signature experiment compares singular-value spectra, row-norm signatures, and inverse- ΔW -L2 signatures at stages 2, 5, 10, and 20, scored against known labels offline. The benchmark uses uniform rank 16 or the repeating heterogeneous schedule (4, 12, 32) with total rank budget 240 across 15 clients. Online discovery and adaptive-rank results are evaluated as separate extensions.

Table 1: Final-round CIFAR-100 accuracy (mean $\pm$ sample standard deviation over seeds 42–44).

Context	Method	Personalized	Consensus
Uniform $r = 16$	Local	57.93 ± 0.55	14.23 ± 1.27
	FedAvg	23.20 ± 0.29	23.20 ± 0.29
	MH	42.30 ± 1.39	15.14 ± 5.10
	Oracle hierarchy	68.98 ± 0.95	10.25 ± 0.62
Heterogeneous 4/12/32, ΔW	Local	53.60 ± 0.62	14.50 ± 0.80
	FedAvg	5.74 ± 0.71	5.73 ± 0.59
	MH	28.45 ± 2.42	6.83 ± 0.45
	Oracle hierarchy	44.07 ± 0.70	11.47 ± 1.13

4 Analysis

For symmetric doubly-stochastic W , with $J = \mathbf{1}\mathbf{1}^\top/N$, consensus error contracts as $\|(W - J)X\|_F \leq |\lambda_2(W)| \|(I - J)X\|_F$. For rank- R matrices, the SVD projection obeys $\|C_r(X) - X\|_F^2 = \sum_{k>r} \sigma_k(X)^2 \leq (1 - r/R) \|X\|_F^2$. With error feedback, the mean update follows the averaged update up to stochastic-gradient terms; without feedback, biased compression leaves an approximation floor. These statements require fixed mixing and bounded residual assumptions; the repository analysis does not claim convergence for the state-dependent discovery mixer or adaptive-rank controller.

5 Experimental Setup

Vision. We use all 50,000 CIFAR-100 training images and 10,000 test images. A frozen ImageNet-pretrained ResNet-18 produces 512-dimensional features after deterministic resize to 224×224 and ImageNet normalization. The trainable classification head is a 100×512 LoRA update. Five superclass groups form domains, with three clients per domain and Dirichlet(0.5) within-domain partitions. Runs use $\alpha = 32$, one local epoch, learning rate 10^{-3} , batch size 128, 50 rounds, and seeds 42, 43, and 44.

Comparators and evidence. We compare local-only, centralized ΔW FedAvg, flat Metropolis–Hastings gossip, oracle two-tier gossip, factor zero-padding, and error feedback. Every run records manifests, split hashes, per-round metrics, per-domain metrics, and communication accounting. The separate Project 2 regeneration uses real CIFAR-10, FashionMNIST, AG News, UCI Heart Disease, and SpeechCommands data; Experiment 1 has 75 client-round rows at seed 42.

Adaptive-rank battery. Project 1 compares fixed rank 32 with the shipped controller ($\gamma = 0.5$) on five real tasks for five rounds at seed 42. Three clients use batch-size capabilities 16, 64, and 256, corresponding to maximum ranks 4, 8, and 16. The controller logs stable-rank demand, EMA state, hysteresis decisions, selected ranks, FLOPs, and capability violations.

6 Results

The oracle hierarchy provides the strongest personalized result in the uniform context, while centralized FedAvg is optimized for a single consensus adapter and is poorly matched to the personalized objective. The heterogeneous context is harder. This known-domain table establishes

Table 2: Project 1 fixed rank 32 versus the shipped adaptive controller after five rounds at seed 42. FLOP savings are measured against the fixed-rank run.

Task	Fixed acc.	Adaptive acc.	Δ acc.	FLOPs saved
CIFAR-CNN	59.50	34.74	-24.76	93.75%
Fashion-MLP	86.55	68.50	-18.05	93.75%
AGNews-LSTM	24.72	26.05	+1.33	93.75%
Tabular-MLP	90.00	83.33	-6.67	93.75%
Audio-1DCNN	96.36	96.36	+0.00	93.75%

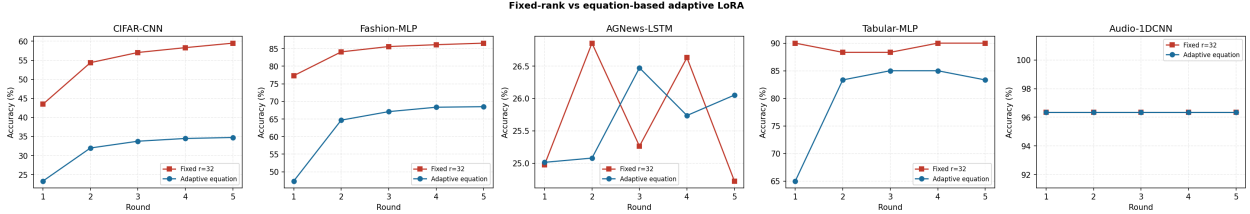


Figure 1: Project 1 fixed-rank versus adaptive-rank accuracy over five rounds on five real tasks.

protocol behavior; the separate online-discovery experiment below evaluates whether domains can be recovered without labels.

The adaptive-rank controller is stable and respects every capability ceiling, but it does not meet the current accuracy-preservation gate when fixed rank 32 is used as the operational compute reference. That reference exceeds the per-client capability maxima (4, 8, and 16), so the reported 93.75% saving is a compute comparison rather than a feasible hardware-matched baseline. The controller selects rank 2 for every client and round (zero rank changes) and falls more than one percentage point below the reference on three tasks. It is therefore an auditable implementation baseline, not a validated accuracy-preserving allocator.

Factor zero-padding under ranks (4, 12, 32) gives personalized accuracy of $4.71 \pm 0.22\%$ for FedAvg, $20.63 \pm 4.40\%$ for MH, and $31.24 \pm 1.27\%$ for oracle hierarchy. The corresponding ΔW oracle result is $44.07 \pm 0.70\%$. Error feedback reaches $26.63 \pm 1.68\%$ for MH and $41.28 \pm 0.71\%$ for oracle hierarchy. These are finite-run comparisons, not universal rankings.

The offline signature screen remains weak for the historical spectral and inverse-distance diagnostics. The new online mixer recovers the five known domain groups on the full CIFAR-100 benchmark: across seeds 42–44 it infers $K = 5$ with confidence 0.72–0.75 and scoring ARI 1.00 at rounds 2, 5, and 10; at round 20 the ARI is 1.00, 0.918, and 1.00, with one seed selecting $K = 6$ and lower confidence. The discovery sketch is 64 floats per client (3,840 fp32 bytes for 15 clients). These labels are used only for scoring; the mixer itself receives adapter states. Because the current runner provides the full state list to the callback, this validates coordinator-visible automatic discovery rather than a neighborhood-only protocol. The regenerated Project 2 calibration selects ridge alpha 1000. Form A has global Spearman correlation 0.368 and pairwise ranking accuracy 0.641; Form B has Spearman 0.369 and pairwise accuracy 0.630. Global R^2 is low (0.079 and 0.025), and task-level correlations change sign across modalities.

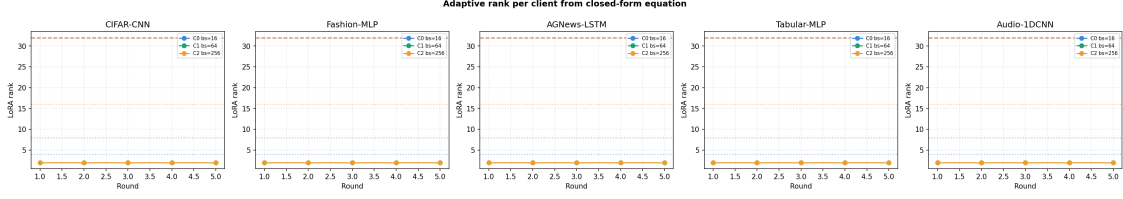


Figure 2: The shipped controller remains at rank 2 for all clients across the five-round battery; capability ceilings are shown as dotted lines.

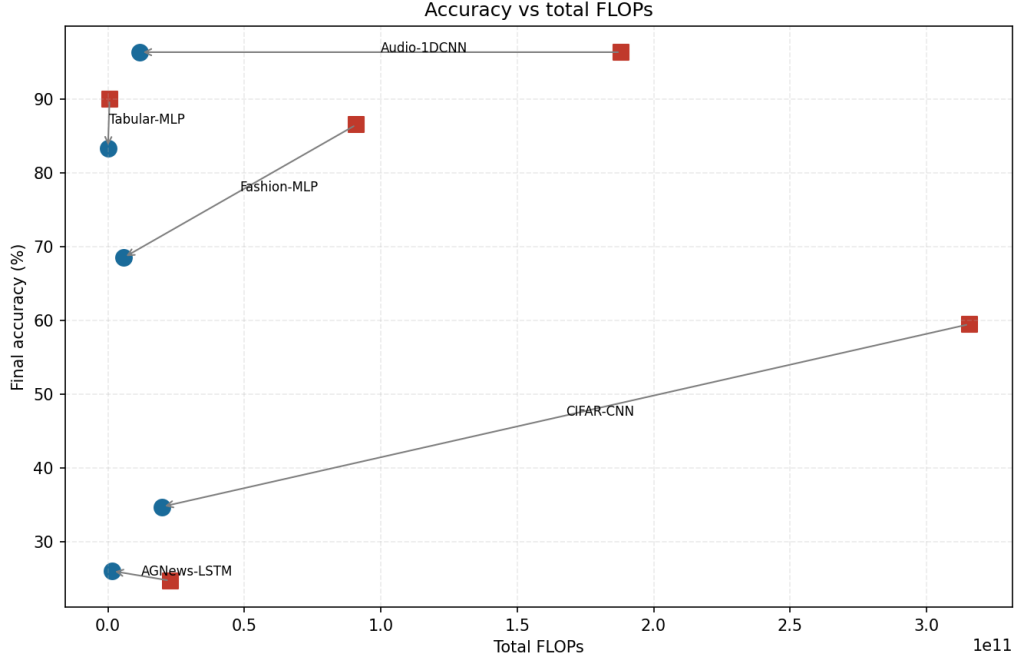


Figure 3: Project 1 accuracy–FLOP trade-off. The adaptive points achieve large compute reductions relative to the fixed rank-32 reference, at the cost of lower accuracy on several tasks.

7 Limitations and Reproducibility

The oracle hierarchy receives true domain assignments, and offline signature scores use those labels only for evaluation. The online discovery result is coordinator-visible because the current runner callback supplies all adapter states; a neighborhood-local discovery protocol and its communication overhead remain future work. The benchmark is one frozen-feature CIFAR-100 task; it does not establish behavior for LLMs, asynchronous churn, or other modalities. Communication is simulated payload accounting rather than measured network throughput. Three seeds provide variability estimates, not statistical proof. P2 calibration is a single-seed, five-round diagnostic with weak predictive fit. Project 1 adaptive-rank calibration is implemented but fails the current fixed-rank-reference accuracy gate: it selects rank 2 throughout, saves 93.75% FLOPs, and loses accuracy on three tasks.

All source code, manifests, compact CSV/JSON evidence, and logs are included. Start with `docs/artifacts/claude_handoff.json`, then read `docs/artifacts/README.md` and the P3 aggregate report. Remote validation used Python 3.10, PyTorch 2.3 with CUDA 12.1, and Tesla V100

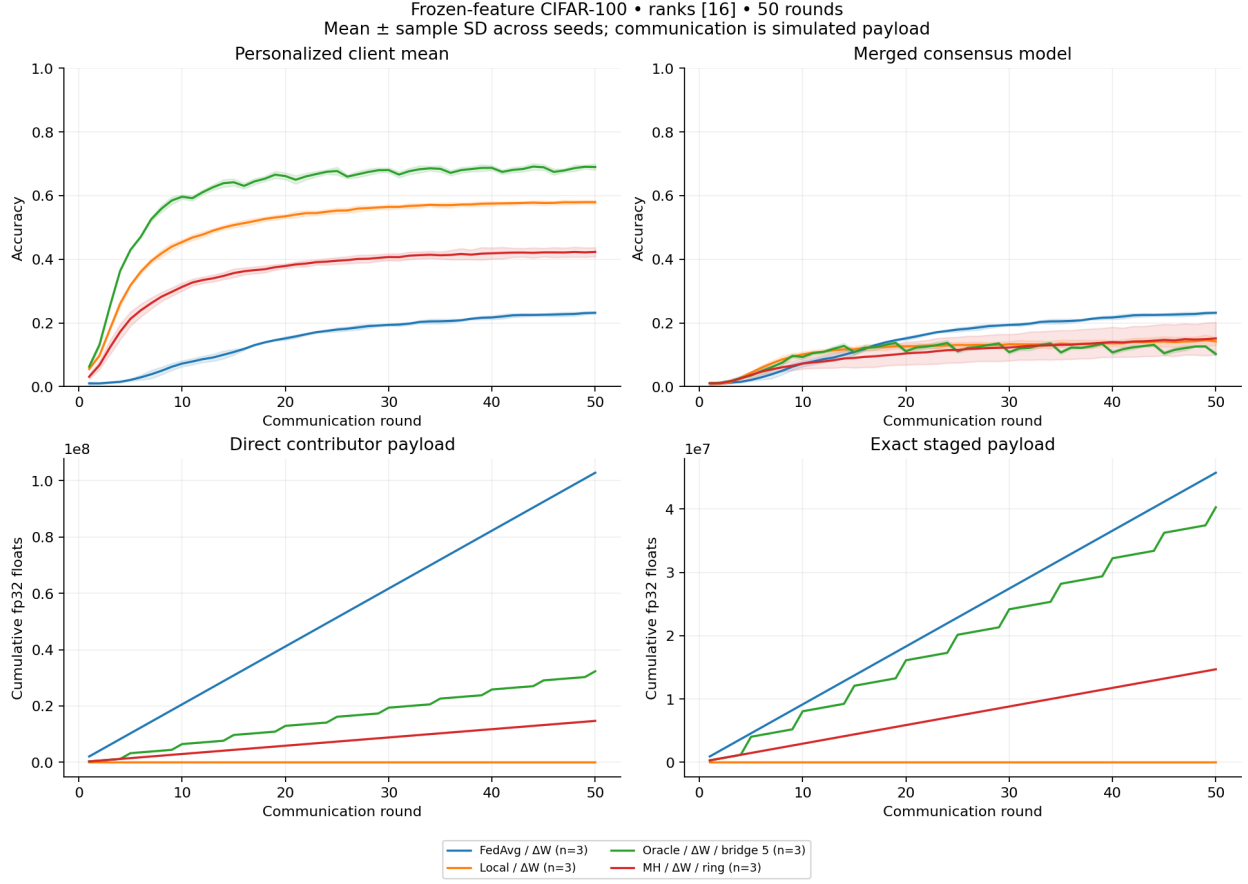


Figure 4: Uniform-rank CIFAR-100 benchmark. Curves show mean $\pm$ sample standard deviation over seeds 42–44. Personalized and consensus accuracy are separate evaluation protocols; payload panels show simulated direct and staged float counts.

GPUs; no credentials are stored in the artifact bundle.

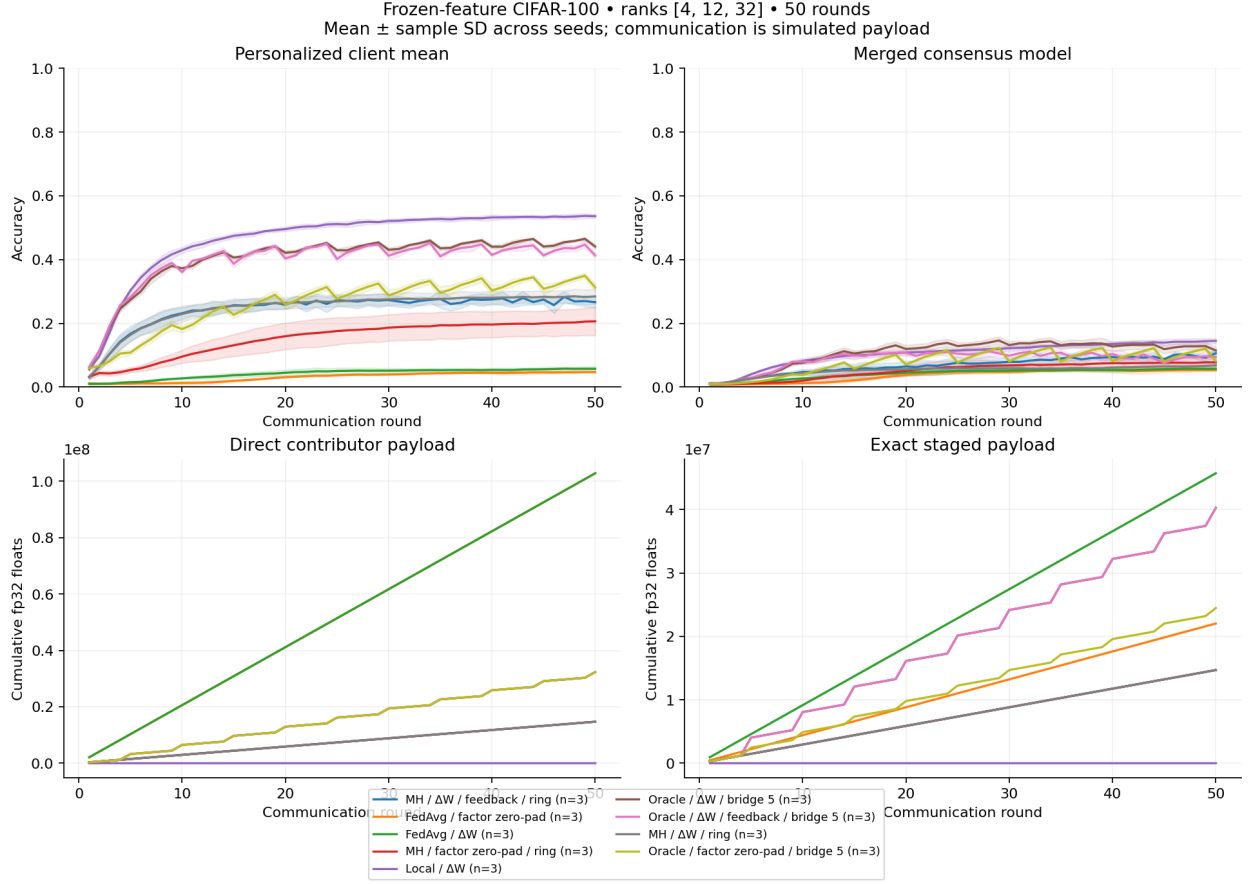


Figure 5: Heterogeneous-rank (4, 12, 32) benchmark. The ΔW and factor-zero-padding arms use the same rank budget but different merge operators; error-feedback results are reported separately in the text and aggregate CSV.

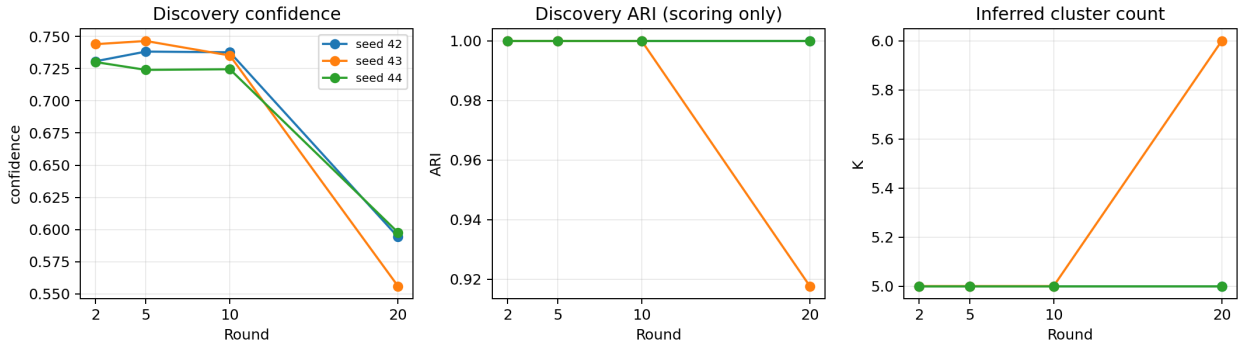


Figure 6: Full-data online domain discovery on CIFAR-100. The EMA/silhouette mixer infers five groups with high confidence through round 10 and near-perfect scoring ARI through round 20; confidence decreases at round 20 and one seed selects $K = 6$.

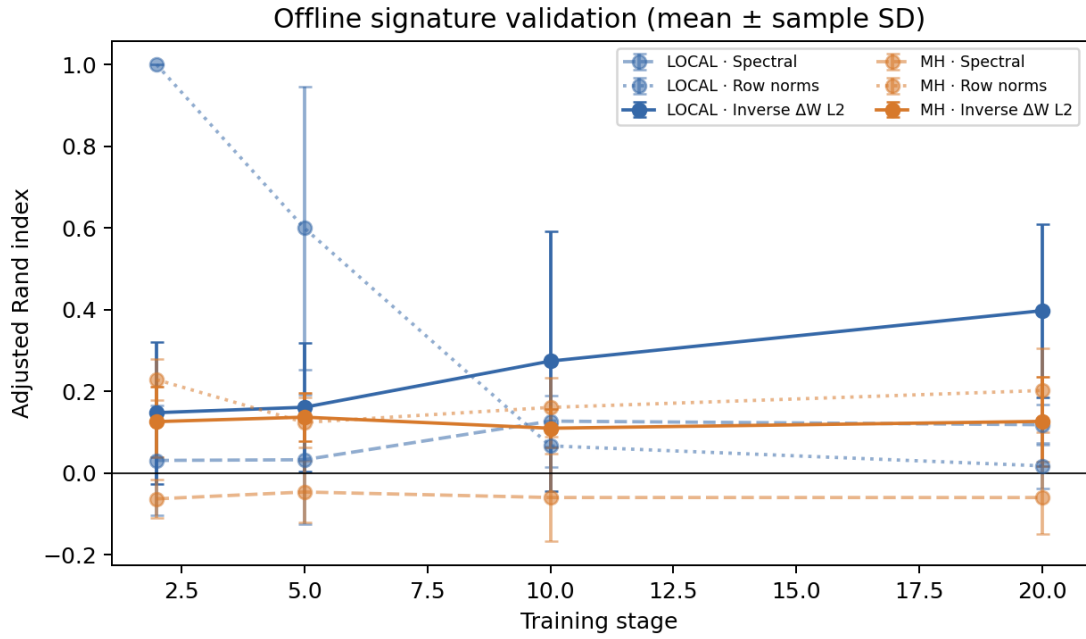


Figure 7: Historical local/MH offline signature validation. The stage-10 screen labels those methods weak-signal; adaptive-run descriptive scores are reported separately with the online discovery result.

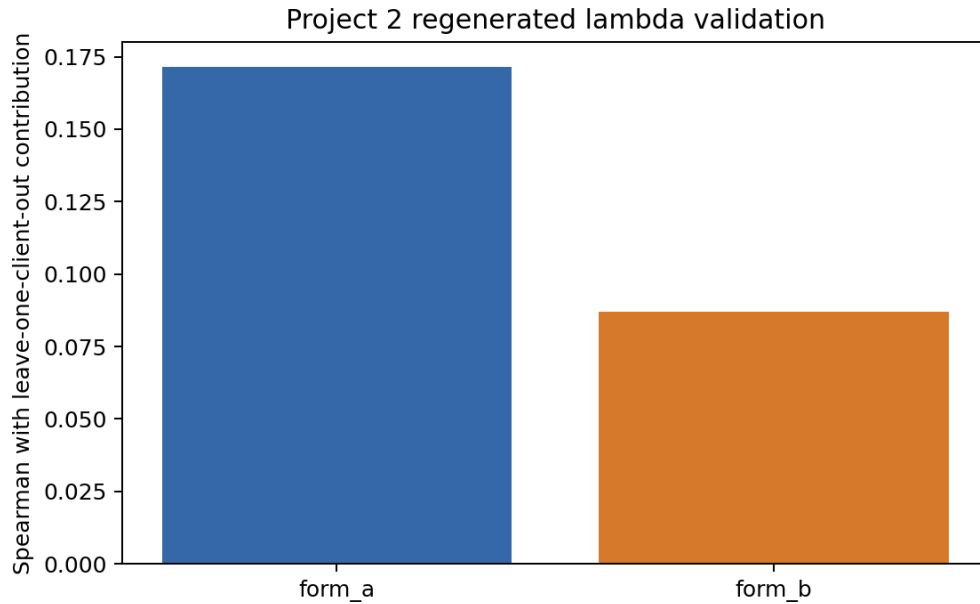


Figure 8: Regenerated Project 2 lambda calibration on 75 real-data observations. Form A and Form B have similar weak global rank association; this calibration is exploratory rather than a validated allocator.

Per-domain diagnostics

The companion domain plots expose the fairness trade-off hidden by a single aggregate accuracy number. They are generated by the same aggregate report and retain the uniform versus heterogeneous protocol split.

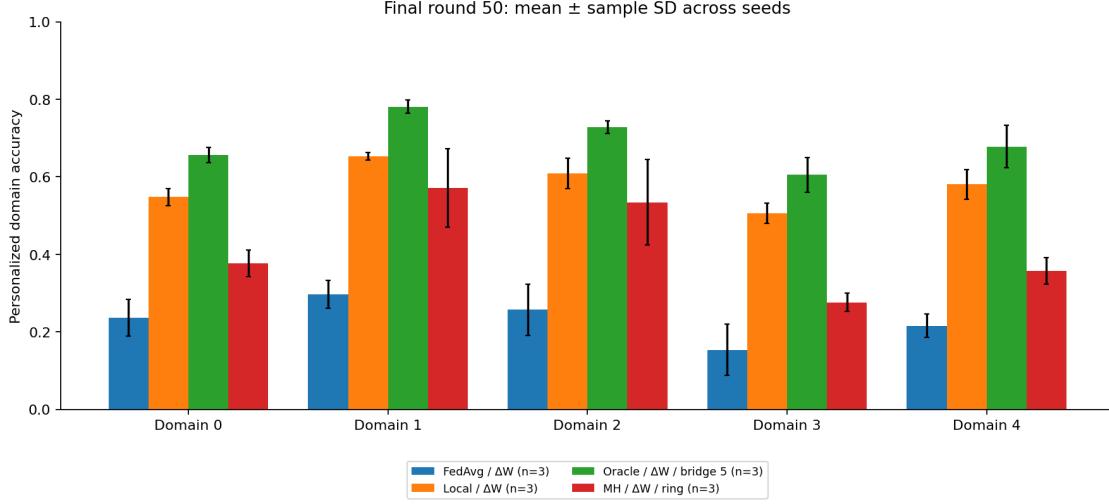


Figure 9: Uniform-rank per-domain personalized accuracy at the final round.

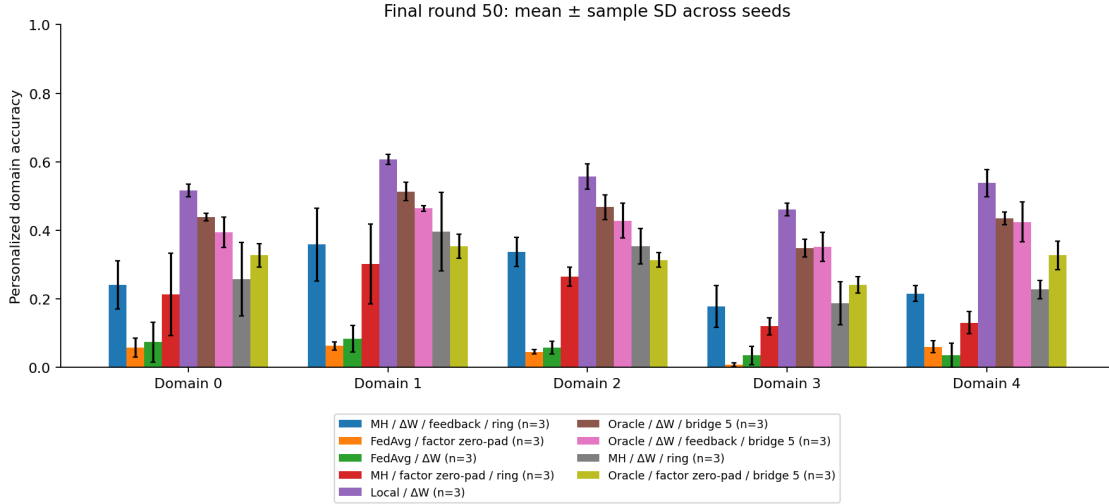


Figure 10: Heterogeneous-rank per-domain personalized accuracy at the final round.

8 Conclusion

AH-LoRA demonstrates a practical primitive for decentralized federated LoRA with heterogeneous ranks: mix scaled effective updates and refactorize locally. The completed benchmark shows that protocol choice and domain structure strongly affect personalized behavior, factor zero-padding can fail under rank heterogeneity, and error feedback changes the compression trade-off. Known-domain hierarchy is a useful oracle comparison, and the online extension demonstrates automatic recovery of the five CIFAR-100 groups under coordinator-visible observations. The adaptive-rank controller is reproducible and FLOP-efficient but fails the current accuracy-preservation gate, exposing calibration as the remaining P1 research problem. The repository provides a complete, reproducible implementation milestone and a concrete base for future work.

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